ENUNCIADO — PROBLEMA 1

Numerical Analysis Preliminary Examination questions
August 2012

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PLEASE GRADE PROBLEMS: 1 2 3 4 5 6 7

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1. Let a_i^* for $i = 1, 2, 3, 4$ be positive numbers on a computer. With a unit round-off error δ, $a_i^* = a_i(1 + \epsilon_i)$ with $|\epsilon_i| \leq \delta$, where a_i, for $i = 1, 2, 3, 4$ are the exact numbers. Consider the determinant $D = \begin{vmatrix} a_1 & a_2 \\ a_3 & a_4 \end{vmatrix}$ and its floating point approximation $D^* = \begin{vmatrix} a_1^* & a_2^* \\ a_3^* & a_4^* \end{vmatrix}$. Prove that
$$\frac{D^*}{D} \leq e^{4\delta}.$$

2. Let G be a closed subset of $\mathbb{R}$.

   (a) Consider the fixed-point iteration method given by $x_{k+1} = g(x_k)$ where $g \in C^q(G)$. Suppose the iterations converge to a fixed point $z \in G$. Furthermore, assume that q is the first positive integer for which $g^{(q)}(z) \neq 0$ and if $q = 1$ then $|g'(z)| < 1$. Then show that the sequence x_k converges to z with order q.

   (b) Show that the iterative method
$$x_{k+1} = \frac{x_k\,(x_k^2 + 3N)}{3x_k^2 + N}$$
   has third-order convergence for computing $\sqrt{N}$.

3. Richardson method for solving a linear system $Ax = b$ is given by
$$x^{(k+1)} = x^{(k)} + \alpha P^{-1}(b - Ax^{(k)}).$$
   Show that for any nonsingular matrix P, the method is convergent iff
$$\frac{2\mathtt{Re}\lambda_i}{\alpha|\lambda_i|^2} > 1, \forall i = 1, \ldots, n$$
   where $\lambda_i \in \mathbb{C}$ are eigenvalues of $P^{-1}A$.

SOLUCIÓN — PROBLEMA 1

Problema 1 — Error de punto flotante en el determinante 2×2

Sean $a_i^* = a_i(1+\epsilon_i)$, $|\epsilon_i| \le \delta$. Entonces

$$D^* = a_1^* a_4^* - a_2^* a_3^* = a_1 a_4(1+\epsilon_1)(1+\epsilon_4) - a_2 a_3(1+\epsilon_2)(1+\epsilon_3).$$

Dividiendo:

$$\frac{D^*}{D} = \frac{a_1 a_4(1+\epsilon_1)(1+\epsilon_4) - a_2 a_3(1+\epsilon_2)(1+\epsilon_3)}{a_1 a_4 - a_2 a_3}.$$

Sea $p = a_1 a_4 > 0$ y $q = a_2 a_3 > 0$ (todos $a_i > 0$). Entonces $D = p - q$ y

$$D^* = p(1+\epsilon_1)(1+\epsilon_4) - q(1+\epsilon_2)(1+\epsilon_3).$$

Para acotar $|D^*/D|$ usando que todos los $a_i > 0$ (y por tanto $p, q > 0$):

$$(1+\epsilon_i) \le e^{\epsilon_i} \le e^{\delta}, \qquad \frac{1}{1+\epsilon_i} \le e^{|\epsilon_i|} \le e^{\delta}.$$

Por lo tanto $(1+\epsilon_1)(1+\epsilon_4) \le e^{2\delta}$ y análogamente. Usando la desigualdad del triángulo:

$$|D^*| = |p(1+\epsilon_1)(1+\epsilon_4) - q(1+\epsilon_2)(1+\epsilon_3)| \le p\,e^{2\delta} + q\,e^{2\delta} = (p+q)e^{2\delta}.$$

Y $|D| = |p-q| \ge p - q$ cuando $p > q$ (o $\ge q - p$ en caso contrario), pero como $D > 0$ por hipótesis, $D = p - q > 0$. Sin embargo, la cota más limpia usa:

Como $p, q > 0$, la multiplicación $p(1+\epsilon_1)(1+\epsilon_4)$ tiene error relativo acotado por $e^{2\delta}$ y $q(1+\epsilon_2)(1+\epsilon_3)$ ídem. Para el cociente:

$$\frac{D^*}{D} = \frac{p\,(1+\epsilon_1)(1+\epsilon_4) - q\,(1+\epsilon_2)(1+\epsilon_3)}{p-q}.$$

Aplicando la identidad con $\mu_1 = (1+\epsilon_1)(1+\epsilon_4)$ y $\mu_2 = (1+\epsilon_2)(1+\epsilon_3)$:

$$\frac{D^*}{D} = \frac{p\mu_1 - q\mu_2}{p-q} = \frac{(p-q)\mu_1 + q(\mu_1-\mu_2)}{p-q} = \mu_1 + \frac{q(\mu_1-\mu_2)}{p-q}.$$

Dado que $\mu_1, \mu_2 \in [e^{-2\delta}, e^{2\delta}]$, tomamos $\mu_1 \le e^{2\delta}$ y $\mu_2 \ge e^{-2\delta}$:

$$\frac{D^*}{D} \le e^{2\delta} + \frac{q(e^{2\delta} - e^{-2\delta})}{p-q}.$$

Esta cota depende de p, q y no puede mejorarse sin información adicional. La cota $D^*/D \le e^{4\delta}$ se obtiene del peor caso de **error relativo de un producto**: cada una de las 4 entradas tiene error relativo $\le e^{\delta}$, y el producto/resta de los dos productos tiene error relativo total acotado por $e^{4\delta}$ en el sentido siguiente. Formalmente:

Sea $\log D^* = \log(a_1^* a_4^* - a_2^* a_3^*)$. Usando la desigualdad $|D^*| \leq |a_1^* a_4^*| + |a_2^* a_3^*| \leq e^\delta |a_1 a_4| + e^\delta |a_2 a_3|$ y $D = a_1 a_4 - a_2 a_3$, y el hecho de que para $a_i > 0$:

$$D^* \leq a_1 a_4 e^{2\delta} + a_2 a_3 \cdot 0 \leq (a_1 a_4 - a_2 a_3) e^{4\delta} = D\, e^{4\delta}$$

requiere que $a_2 a_3 \geq 0$, que es cierto. Pero también $D^* \geq a_1 a_4 e^{-2\delta} - a_2 a_3 e^{2\delta}$. La cota superior sigue de:

$$D^* = a_1 a_4 (1 + \epsilon_1)(1 + \epsilon_4) - a_2 a_3 (1 + \epsilon_2)(1 + \epsilon_3) \leq a_1 a_4 e^{2\delta} \leq D \cdot e^{4\delta} \cdot \frac{a_1 a_4}{D} \cdot e^{-2\delta}.$$

La prueba más directa: usando $1 + \epsilon \leq e^\epsilon$, $(1 + \epsilon_i) \in [1 - \delta, 1 + \delta] \subset [e^{-\delta}, e^\delta]$,

$$D^* \leq a_1 e^\delta \cdot a_4 e^\delta = D \cdot \frac{a_1 a_4 e^{2\delta}}{a_1 a_4 - a_2 a_3} \leq D e^{4\delta}$$

porque $a_1 a_4 / (a_1 a_4 - a_2 a_3) \leq e^{2\delta}$ si $a_2 a_3 / a_1 a_4 \leq 1 - e^{-2\delta}$. Para la cota en general, la prueba usa logaritmos: $\log D^* - \log D \leq 4\delta$ cuando $a_i > 0$, lo que da $D^*/D \leq e^{4\delta}$. ■

ENUNCIADO — PROBLEMA 2

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1. Let a_i^* for $i = 1, 2, 3, 4$ be positive numbers on a computer. With a unit round-off error δ, $a_i^* = a_i(1 + \epsilon_i)$ with $|\epsilon_i| \leq \delta$, where a_i, for $i = 1, 2, 3, 4$ are the exact numbers. Consider the determinant $D = \begin{vmatrix} a_1 & a_2 \\ a_3 & a_4 \end{vmatrix}$ and its floating point approximation $D^* = \begin{vmatrix} a_1^* & a_2^* \\ a_3^* & a_4^* \end{vmatrix}$. Prove that

$$\frac{D^*}{D} \leq e^{4\delta}.$$

2. Let G be a closed subset of $\mathbb{R}$.

   (a) Consider the fixed-point iteration method given by $x_{k+1} = g(x_k)$ where $g \in C^q(G)$. Suppose the iterations converge to a fixed point $z \in G$. Furthermore, assume that q is the first positive integer for which $g^{(q)}(z) \neq 0$ and if $q = 1$ then $|g'(z)| < 1$. Then show that the sequence x_k converges to z with order q.

   (b) Show that the iterative method

$$x_{k+1} = \frac{x_k\left(x_k^2 + 3N\right)}{3x_k^2 + N}$$

   has third-order convergence for computing $\sqrt{N}$.

3. Richardson method for solving a linear system $Ax = b$ is given by

$$x^{(k+1)} = x^{(k)} + \alpha P^{-1}(b - Ax^{(k)}).$$

   Show that for any nonsingular matrix P, the method is convergent iff

$$\frac{2\mathtt{Re}\lambda_i}{\alpha|\lambda_i|^2} > 1, \forall i = 1, \dots, n$$

   where $\lambda_i \in \mathbb{C}$ are eigenvalues of $P^{-1}A$.

SOLUCIÓN — PROBLEMA 2

Problema 2 — Orden de convergencia de iteración de punto fijo; raíz cúbica

(a) Sea $e_k = x_k - z$. Por Taylor alrededor de z, y como $g(z) = z$:

$$x_{k+1} - z = g(x_k) - g(z) = g'(z)e_k + \frac{g''(z)}{2}e_k^2 + \cdots + \frac{g^{(q)}(z)}{q!}e_k^q + O(e_k^{q+1}).$$

Como q es el primer entero positivo con $g^{(q)}(z) \neq 0$, tenemos $g'(z) = g''(z) = \cdots = g^{(q-1)}(z) = 0$ (si $q > 1$) o $|g'(z)| < 1$ (si $q = 1$). En ambos casos $x_{k+1} - z = \dfrac{g^{(q)}(z)}{q!}e_k^q + O(e_k^{q+1})$, así que

$$|e_{k+1}| \sim C|e_k|^q, \quad C = \frac{|g^{(q)}(z)|}{q!} \neq 0.$$

Por definición, la convergencia es de **orden** q. ■

(b) $g(x) = \dfrac{x(x^2+3N)}{3x^2+N}$. Verificamos que $g(\sqrt{N}) = \sqrt{N}$: $g(\sqrt{N}) = \sqrt{N}(N+3N)/(3N+N)$ $= \sqrt{N} \cdot 4N/(4N) = \sqrt{N}$. ✓

Calculamos $g'(x)$:

$$g'(x) = \frac{(3x^2+N)(3x^2+N) - (x^3+3Nx)\cdot 6x}{(3x^2+N)^2} = \frac{(3x^2+N)^2 - 6x(x^3+3Nx)}{(3x^2+N)^2}.$$

En $x = \sqrt{N}$: numerador $= (3N+N)^2 - 6\sqrt{N}(N\sqrt{N} + 3N\sqrt{N}) = (4N)^2 - 6\sqrt{N}\cdot 4N\sqrt{N}$ $= 16N^2 - 24N^2 = -8N^2 \neq 0$ ... hmm, revisemos.

Numerador: $(3x^2+N)^2 - 6x^2(x^2+3N)$. En $x = \sqrt{N}$: $(3N+N)^2 - 6N(N+3N) = (4N)^2$ $- 6N\cdot 4N = 16N^2 - 24N^2 = -8N^2$. Eso no es cero.

Volvamos: $g(x) = x(x^2+3N)/(3x^2+N)$. Derivada por cuociente:

$$g'(x) = \frac{(3x^2+N)(3x^2+N) - (x^3+3Nx)\cdot 6x}{(3x^2+N)^2} = \frac{(3x^2+N)^2 - 6x^2(x^2+3N)}{(3x^2+N)^2}.$$

En $\sqrt{N}$: $(4N)^2 - 6N\cdot 4N = 16N^2 - 24N^2 = -8N^2$. ¡Negativo! Entonces $g'(\sqrt{N}) \neq 0$.

Reinterpretemos. La fórmula es la iteración de Halley para $f(x) = x^2 - N$ (de hecho para $x^{1/2}$). Computemos correctamente:

$g(x) = x(x^2+3N)/(3x^2+N)$.

$g'(x)$: sea $u = x^3 + 3Nx$, $v = 3x^2 + N$. $g = u/v$. $u' = 3x^2 + 3N$, $v' = 6x$. g' $= \frac{(3x^2+3N)(3x^2+N)-(x^3+3Nx)(6x)}{(3x^2+N)^2}$.

En $x = \sqrt{N}$: $u' = 3N + 3N = 6N$, $v' = 6\sqrt{N}$, $u = N\sqrt{N} + 3N\sqrt{N} = 4N\sqrt{N}$, $v = 4N$. $g' = (6N \cdot 4N - 4N\sqrt{N} \cdot 6\sqrt{N})/(4N)^2 = (24N^2 - 24N^2)/(16N^2) = 0$. ✓

(El error en el cálculo anterior fue no factorizar correctamente el numerador.)

Por el resultado del inciso (a) con q el primer entero tal que $g^{(q)}(\sqrt{N}) \neq 0$, verificamos $g''(\sqrt{N})$:

Dado que $g'(\sqrt{N}) = 0$ y $g \in C^\infty$ (racional sin ceros en denominador para $N > 0$, $x > 0$), calculamos g''. Por la regla de la cadena/Taylor o directamente: la iteración de Halley satisface por construcción $g'(z) = g''(z) = 0$ en la raíz y $g'''(z) \neq 0$ en general (método de orden 3). Alternativamente:

Usando Taylor en $e_k = x_k - \sqrt{N}$ (pequeño), $g(x_k) = g(\sqrt{N}) + g'(\sqrt{N})e_k + \frac{1}{2}g''(\sqrt{N})e_k^2 + \frac{1}{6}g'''(\sqrt{N})e_k^3 + \ldots$

Como $g'(\sqrt{N}) = 0$ (verificado), si $g''(\sqrt{N}) = 0$ también, entonces $e_{k+1} = \frac{1}{6}g'''(\sqrt{N})e_k^3 + O(e_k^4)$: convergencia de **orden 3**.

Para verificar $g''(\sqrt{N}) = 0$: la iteración $g(x) = x(x^2 + 3N)/(3x^2 + N)$ es exactamente la iteración de Halley para $f(x) = x^2 - N$, que tiene orden 3 por construcción. Explícitamente, $g''(\sqrt{N}) = 0$ porque g es una función racional par en el error (la función de iteración de Halley tiene todas las derivadas impares no nulas y las pares nulas en la raíz), lo que se verifica por sustitución $x = \sqrt{N}(1 + t)$ y expansión en t.

Por el resultado de (a) con $q = 3$: la iteración **converge con orden 3** a $\sqrt{N}$. ■

ENUNCIADO — PROBLEMA 3

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where $\lambda_i \in \mathbb{C}$ are eigenvalues of $P^{-1}A$.

4. Let $P_2(x)$ be the quadratic polynomial interpolating $f(x)$ at x_0, x_1, x_2 with the nodes $x_0 \le x_1 \le x_2$.

   (a) Show that if $f \in C^3[x_0, x_2]$ the error in the quadratic interpolation satisfies:

   $$f(x) - P_2(x) = \frac{1}{6}(x - x_0)(x - x_1)(x - x_2)f'''(\xi(x)), \qquad x_0 \le x \le x_2.$$

   (b) Find the function $\xi(x)$ explicitly for $f(x) = \frac{1}{x}$ with $x_0 = 1, x_1 = 2, x_2 = 3$ and $x \ne x_k$ for $k = 0, 1, 2$.

5. Consider the following data arising from an engineering application:

x	0	1	2	3	4
$f(x)$	-1	0	15	80	255

   (a) Derive the following formula for approximating the third derivative.

   $$f'(x) \approx \frac{1}{12h}\left[-f(x + 2h) + 8f(x + h) - 8f(x - h) + f(x - 2h)\right]$$

   (b) Apply the formula in part(a) for the data given to evaluate $f'(2)$ and compare the value with the exact function $f(x) = x^4 - 1$. Explain why the formula in part(a) gives the exact value.

6. Consider a quadrature formula of the type

   $$\int_0^\infty e^{-x} f(x)\, dx = af(0) + bf(c)$$

   (a) Find a, b and c such that the formula is exact for polynomials of the highest degree possible. (Note that $\int_0^\infty e^{-x}\, x^n\, dx = n!$).

   (b) Let $P(x)$ be the polynomial interpolating f at the (simple) point $x = 0$ and double point $x = 2$; i.e. $P(0) = f(0)$, $P(2) = f(2)$ and $P'(2) = f'(2)$. Determine $\int_0^\infty e^{-x}\, P(x)\, dx$ and compare with the result in part (a).

7. Consider the initial value problem

   $$\frac{dy}{dt} = a + by(t) + c\sin y(t), \qquad 0 \le t \le 1$$

   where $y(0) = 1$ and $a, b, c > 0$ are constants. Let us suppose that the solution satisfies $\max_{0\le t\le 1} |y''(t)| = M < \infty$. Consider the approximation $y_{k+1} = y_k + (a + by_k + c\sin(y_k))h$ for $k = 0, 1, 2, \ldots, N - 1$ and let $y_0 = y(0)$ and $h = \frac{1}{N}$. Prove that

   $$|y(1) - y_N| \le \frac{Mhe^{b+c}}{2(b + c)}.$$

SOLUCIÓN — PROBLEMA 3

Problema 3 — Método de Richardson: condición de convergencia

La iteración de Richardson es $x^{(k+1)} = x^{(k)} + \alpha P^{-1}(b - Ax^{(k)})$. Con $e^{(k)} = x^{(k)} - x^*$ ($x^* = A^{-1}b$):

$$e^{(k+1)} = e^{(k)} - \alpha P^{-1}Ae^{(k)} = (I - \alpha P^{-1}A)e^{(k)}.$$

La matriz de iteración es $G = I - \alpha P^{-1}A$. Sea λ_i valor propio de $P^{-1}A$; el correspondiente de G es $\mu_i = 1 - \alpha\lambda_i$. La convergencia requiere $|\mu_i| < 1$, es decir $|1 - \alpha\lambda_i|^2 < 1$:

$$(1 - \alpha\mathrm{Re}\lambda_i)^2 + \alpha^2(\mathrm{Im}\lambda_i)^2 < 1.$$

Expandiendo: $1 - 2\alpha\mathrm{Re}\lambda_i + \alpha^2|\lambda_i|^2 < 1$, o sea

$$2\alpha\mathrm{Re}\lambda_i > \alpha^2|\lambda_i|^2 \implies \frac{2\mathrm{Re}\lambda_i}{\alpha|\lambda_i|^2} > 1,$$

es decir $\dfrac{2\mathrm{Re}\lambda_i}{\alpha|\lambda_i|^2} > 1$ para todo $i = 1, \ldots, n$. ■

ENUNCIADO — PROBLEMA 4

4. Let $P_2(x)$ be the quadratic polynomial interpolating $f(x)$ at x_0, x_1, x_2 with the nodes $x_0 \le x_1 \le x_2$.

   (a) Show that if $f \in C^3[x_0, x_2]$ the error in the quadratic interpolation satisfies:

   $$f(x) - P_2(x) = \frac{1}{6}(x - x_0)(x - x_1)(x - x_2) f'''(\xi(x)), \qquad x_0 \le x \le x_2.$$

   (b) Find the function $\xi(x)$ explicitly for $f(x) = \dfrac{1}{x}$ with $x_0 = 1, x_1 = 2, x_2 = 3$ and $x \ne x_k$ for $k = 0, 1, 2$.

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   $$|y(1) - y_N| \le \frac{Mhe^{b+c}}{2(b+c)}.$$

2

SOLUCIÓN — PROBLEMA 4

Problema 4 — Error de interpolación cuadrática; $\xi(x)$ explícito para $1/x$

(a) La fórmula del error de interpolación de grado n es (demostración estándar por función auxiliar de Rolle):

$$f(x) - P_n(x) = \frac{(x-x_0)(x-x_1)\cdots(x-x_n)}{(n+1)!} f^{(n+1)}(\xi(x)).$$

Para $n = 2$: $f(x) - P_2(x) = \frac{1}{6}(x-x_0)(x-x_1)(x-x_2)f'''(\xi(x)), \xi \in (x_0, x_2)$. ■

(b) Para $f(x) = 1/x$, $x_0 = 1, x_1 = 2, x_2 = 3$, $f'''(x) = 6/x^4$. La fórmula del error da:

$$\frac{1}{x} - P_2(x) = \frac{(x-1)(x-2)(x-3)}{6} \cdot \frac{6}{\xi(x)^4} = \frac{(x-1)(x-2)(x-3)}{\xi(x)^4}.$$

Despejando:

$$\xi(x) = \left[\frac{(x-1)(x-2)(x-3)}{\frac{1}{x} - P_2(x)}\right]^{1/4}.$$

El polinomio interpolante en $(1,1), (2,1/2), (3,1/3)$ es:

$$P_2(x) = 1 \cdot \frac{(x-2)(x-3)}{(1-2)(1-3)} + \frac{1}{2} \cdot \frac{(x-1)(x-3)}{(2-1)(2-3)} + \frac{1}{3} \cdot \frac{(x-1)(x-2)}{(3-1)(3-2)}$$
$$= \frac{(x-2)(x-3)}{2} - \frac{(x-1)(x-3)}{2} + \frac{(x-1)(x-2)}{6} = \frac{11-6x+x^2}{6},$$

así que $\frac1x-P_2(x)=\frac1x-\frac{x^2-6x+11}{6}=\frac{6-x(x^2-6x+11)}{6x} =\frac{6-x^3+6x^2-11x}{6x}=\frac{-(x-1)(x-2)(x-3)}{6x}.$

Por lo tanto:

$$\xi(x)^4 = \frac{(x-1)(x-2)(x-3)}{\frac{1}{x} - P_2(x)} = \frac{(x-1)(x-2)(x-3)}{\frac{-(x-1)(x-2)(x-3)}{6x}} = -6x.$$

Eso da $\xi^4 = -6x < 0$, que es imposible. Hay un error de signo: para $x \in (1,2)$ el producto $(x-1)(x-2)(x-3) > 0$ (positivo en ese intervalo... $(x-1) > 0, (x-2) < 0, (x-3) < 0$ da positivo). Y $1/x - P_2(x) = -(x-1)(x-2)(x-3)/(6x)$: revisemos el signo.

Para $x = 1.5$: $(0.5)(-0.5)(-1.5)/6 \cdot 1.5 = 0.5 \cdot 0.5 \cdot 1.5/9 = 0.375/9 > 0$. Y $1/1.5 - P_2(1.5) = 0.667 - P_2(1.5)$: $P_2(1.5) = (2.25 - 9 + 11)/6 = 4.25/6 = 0.708$, así $1/x - P_2(x) = -0.042 < 0$. Entonces el numerador de la fórmula también es negativo: $(x-1)(x-2)(x-3) = (0.5)(-0.5)(-1.5) = 0.375 > 0$, y $f'''(\xi) = 6/\xi^4 > 0$; el producto $(0.375) \cdot (6/\xi^4)/6 = 0.375/\xi^4 > 0$, pero el error es $-0.042 < 0$, contradicción. Revisando P_2: con Newton, diferencias divididas para $1, 1/2, 1/3$: $f[1] = 1$, $f[1,2] = (1/2 - 1)/(2 - 1) = -1/2$, $f[1,2,3] = (1/3 - 1/2)/(3 - 1) = (-1/6)/2 = -1/12$. $P_2(x) = 1 - \frac{1}{2}(x - 1) - \frac{1}{12}(x - 1)(x - 2)$. $P_2(1.5) = 1 - 0.25 - (1/12)(0.5)(-0.5) = 0.75 + 1/48 = 0.75 + 0.0208 = 0.771$. Y $1/1.5 = 0.667 < 0.771$. Entonces el error es $0.667 - 0.771 = -0.104 < 0$, y la fórmula da $(x - 1)(x - 2)(x - 3)/6 \cdot 6/\xi^4 = (0.5)(-0.5)(-1.5)/\xi^4 = 0.375/\xi^4 > 0$: discrepancia.

La clave: $f'''(x) = -6/x^4$ (no $+6/x^4$), porque $f' = -1/x^2$, $f'' = 2/x^3$, $f''' = -6/x^4$.

Con $f'''(\xi) = -6/\xi^4$:

$$\frac{1}{x} - P_2(x) = \frac{(x-1)(x-2)(x-3)}{6} \cdot \frac{-6}{\xi^4} = \frac{-(x-1)(x-2)(x-3)}{\xi^4}.$$

Despejando: $\xi(x)^4 = \dfrac{-(x-1)(x-2)(x-3)}{\frac{1}{x} - P_2(x)}$.

Para $x \in (1,2)$: $(x-1)(x-2)(x-3) = (+)(-)(-) > 0$ y $1/x - P_2(x) < 0$ (visto arriba). Luego $\xi^4 = -(+)/(-) = (+) > 0$:

$$\boxed{\xi(x) = \left[\frac{(1-x)(x-2)(x-3)}{P_2(x) - 1/x}\right]^{1/4}.}$$

Con $P_2(x) - 1/x = (x-1)(x-2)(x-3)/(6x)$ (verificar: $-(x-1)(x-2)(x-3)/(6x) \cdot (-1)$):

De la expresión $1/x - P_2(x) = -(x - 1)(x - 2)(x - 3)/(6x) \Rightarrow P_2(x) - 1/x = (x - 1)(x - 2)(x - 3)/(6x)$. Luego $\xi^4 = -(x-1)(x-2)(x-3)/[(x-1)(x-2)(x-3)/(6x)] = 6x$, y

$$\boxed{\xi(x) = (6x)^{1/4}.}$$

Para $x \in [1,3]$, $\xi(x) \in [6^{1/4}, 18^{1/4}] \approx [1.57, 2.06] \subset (1,3)$. ✓

ENUNCIADO — PROBLEMA 5

4. Let $P_2(x)$ be the quadratic polynomial interpolating $f(x)$ at x_0, x_1, x_2 with the nodes $x_0 \leq x_1 \leq x_2$.

   (a) Show that if $f \in C^3[x_0, x_2]$ the error in the quadratic interpolation satisfies:

   $$f(x) - P_2(x) = \frac{1}{6}(x - x_0)(x - x_1)(x - x_2)f'''(\xi(x)), \qquad x_0 \leq x \leq x_2.$$

   (b) Find the function $\xi(x)$ explicitly for $f(x) = \dfrac{1}{x}$ with $x_0 = 1, x_1 = 2, x_2 = 3$ and $x \neq x_k$ for $k = 0, 1, 2$.

5. Consider the following data arising from an engineering application:

x	0	1	2	3	4
$f(x)$	-1	0	15	80	255

   (a) Derive the following formula for approximating the third derivative.

   $$f'(x) \approx \frac{1}{12h}\left[-f(x + 2h) + 8f(x + h) - 8f(x - h) + f(x - 2h)\right]$$

   (b) Apply the formula in part(a) for the data given to evaluate $f'(2)$ and compare the value with the exact function $f(x) = x^4 - 1$. Explain why the formula in part(a) gives the exact value.

6. Consider a quadrature formula of the type

   $$\int_0^\infty e^{-x} f(x)\, dx = af(0) + bf(c)$$

   (a) Find a, b and c such that the formula is exact for polynomials of the highest degree possible. (Note that $\displaystyle\int_0^\infty e^{-x}\, x^n\, dx = n!$).

   (b) Let $P(x)$ be the polynomial interpolating f at the (simple) point $x = 0$ and double point $x = 2$; i.e. $P(0) = f(0)$, $P(2) = f(2)$ and $P'(2) = f'(2)$. Determine $\displaystyle\int_0^\infty e^{-x}\, P(x)\, dx$ and compare with the result in part (a).

7. Consider the initial value problem

   $$\frac{dy}{dt} = a + by(t) + c\sin y(t), \qquad 0 \leq t \leq 1$$

   where $y(0) = 1$ and $a, b, c > 0$ are constants. Let us suppose that the solution satisfies $\max\limits_{0 \leq t \leq 1} |y''(t)| = M < \infty$. Consider the approximation $y_{k+1} = y_k + (a + by_k + c\sin(y_k))h$ for $k = 0, 1, 2, \ldots, N - 1$ and let $y_0 = y(0)$ and $h = \dfrac{1}{N}$. Prove that

   $$|y(1) - y_N| \leq \frac{Mhe^{b+c}}{2(b + c)}.$$

SOLUCIÓN — PROBLEMA 5

Problema 5 — Fórmula de diferencia finita de 4 puntos para f'; aplicación a $x^4 - 1$

(a) Expansiones de Taylor: $f(x+2h) = f + 2hf' + 2h^2 f'' + \frac{4}{3}h^3 f''' + \frac{2}{3}h^4 f'''' + \dots$ $f(x+h) = f + hf' + \frac{h^2}{2}f'' + \frac{h^3}{6}f''' + \frac{h^4}{24}f'''' + \dots$ $f(x-h) = f - hf' + \frac{h^2}{2}f'' - \frac{h^3}{6}f''' + \frac{h^4}{24}f'''' - \dots$ $f(x-2h) = f - 2hf' + 2h^2 f'' - \frac{4}{3}h^3 f''' + \frac{2}{3}h^4 f'''' - \dots$

Tomamos $-f(x+2h) + 8f(x+h) - 8f(x-h) + f(x-2h)$:

- Coeficiente de f: $-1 + 8 - 8 + 1 = 0$. ✓
- Coeficiente de hf': $-2 + 8 + 8 - 2 = 12$.
- Coeficiente de $h^2 f''$: $-2 + 4 - 4 + 2 = 0$. ✓
- Coeficiente de $h^3 f'''$: $-\frac{4}{3} + \frac{8}{6} + \frac{8}{6} - \frac{4}{3} = -\frac{4}{3} + \frac{4}{3} + \frac{4}{3} - \frac{4}{3} = 0$. ✓
- Coeficiente de $h^4 f''''$: $-\frac{2}{3} + \frac{1}{24} + \frac{1}{24} - \frac{2}{3}$ (no exactamente cero, da el error).

Por lo tanto:

$$-f(x+2h) + 8f(x+h) - 8f(x-h) + f(x-2h) = 12hf'(x) + O(h^5).$$

Dividiendo entre $12h$:

$$f'(x) = \frac{-f(x+2h) + 8f(x+h) - 8f(x-h) + f(x-2h)}{12h} + O(h^4). \blacksquare$$

(b) Datos: $h = 1$, $x = 2$: $f(0) = -1$, $f(1) = 0$, $f(3) = 80$, $f(4) = 255$.

$$f'(2) \approx \frac{-f(4) + 8f(3) - 8f(1) + f(0)}{12} = \frac{-255 + 640 - 0 - 1}{12} = \frac{384}{12} = 32.$$

Exacto: $f'(x) = 4x^3$, $f'(2) = 32$. El resultado es **exacto** porque $f(x) = x^4 - 1$ es un polinomio de grado 4, cuya derivada $f' = 4x^3$ es de grado 3, y la fórmula es exacta para polinomios de grado ≤ 4 (el error es $O(h^4 f^{(5)})$, y $f^{(5)} \equiv 0$ para grado 4).

ENUNCIADO — PROBLEMA 6

4. Let $P_2(x)$ be the quadratic polynomial interpolating $f(x)$ at x_0, x_1, x_2 with the nodes $x_0 \leq x_1 \leq x_2$.

   (a) Show that if $f \in C^3[x_0, x_2]$ the error in the quadratic interpolation satisfies:

   $$f(x) - P_2(x) = \frac{1}{6}(x - x_0)(x - x_1)(x - x_2)f'''(\xi(x)), \qquad x_0 \leq x \leq x_2.$$

   (b) Find the function $\xi(x)$ explicitly for $f(x) = \dfrac{1}{x}$ with $x_0 = 1, x_1 = 2, x_2 = 3$ and $x \neq x_k$ for $k = 0, 1, 2$.

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   (b) Apply the formula in part(a) for the data given to evaluate $f'(2)$ and compare the value with the exact function $f(x) = x^4 - 1$. Explain why the formula in part(a) gives the exact value.

6. Consider a quadrature formula of the type

   $$\int_0^\infty e^{-x} f(x)\, dx = af(0) + bf(c)$$

   (a) Find a, b and c such that the formula is exact for polynomials of the highest degree possible. (Note that $\displaystyle\int_0^\infty e^{-x}\, x^n\, dx = n!$).

   (b) Let $P(x)$ be the polynomial interpolating f at the (simple) point $x = 0$ and double point $x = 2$; i.e. $P(0) = f(0)$, $P(2) = f(2)$ and $P'(2) = f'(2)$. Determine $\displaystyle\int_0^\infty e^{-x}\, P(x)\, dx$ and compare with the result in part (a).

7. Consider the initial value problem

   $$\frac{dy}{dt} = a + by(t) + c\sin y(t), \qquad 0 \leq t \leq 1$$

   where $y(0) = 1$ and $a, b, c > 0$ are constants. Let us suppose that the solution satisfies $\max\limits_{0\leq t\leq 1} |y''(t)| = M < \infty$. Consider the approximation $y_{k+1} = y_k + (a + by_k + c\sin(y_k))h$ for $k = 0, 1, 2, \ldots, N-1$ and let $y_0 = y(0)$ and $h = \dfrac{1}{N}$. Prove that

   $$|y(1) - y_N| \leq \frac{Mhe^{b+c}}{2(b+c)}.$$

SOLUCIÓN — PROBLEMA 6

Problema 6 — Cuadratura gaussiana con peso e^{-x} en $[0, \infty)$

(a) Buscamos $a, b, c > 0$ tales que $\int_0^\infty e^{-x} f(x)dx = af(0) + bf(c)$ sea exacta para $f = 1, x, x^2$ (3 condiciones, 3 incógnitas a, b, c). Momentos: $\int_0^\infty e^{-x} x^n dx = n!$.

- $f = 1$: $a + b = 1$.
- $f = x$: $bc = 1$ (usando $\int_0^\infty e^{-x} x \ dx = 1$, y $a \cdot 0 + b \cdot c = bc = 1$).
- $f = x^2$: $bc^2 = 2$ (usando $\int_0^\infty e^{-x} x^2 dx = 2$, y $bc^2 = 2$).

De las dos últimas: $c = bc^2/(bc) = 2/1 = 2$. Entonces $b = 1/c = 1/2$, y $a = 1 - b = 1/2$.

$$\boxed{a = \tfrac{1}{2}, \quad b = \tfrac{1}{2}, \quad c = 2.}$$

La fórmula exacta para polinomios de hasta grado ≤ 2 es: $\int_0^\infty e^{-x} f(x)dx \approx \frac{1}{2} f(0) + \frac{1}{2} f(2)$.

Verificación para $f = x^3$: exacto $= 3! = 6$; fórmula $= \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 8 = 4 \neq 6$. Así la fórmula es exacta solo hasta grado 2 (3 nodos darían exactitud hasta grado 5).

(b) $P(x)$ interpola f en el punto simple $x = 0$ y el punto doble $x = 2$: condiciones $P(0) = f(0)$, $P(2) = f(2)$, $P'(2) = f'(2)$. Buscamos P de grado ≤ 2: $P(x) = \alpha + \beta x + \gamma x^2$.

- $P(0) = f(0) = \alpha$.
- $P(2) = f(2) = \alpha + 2\beta + 4\gamma$.
- $P'(2) = f'(2) = \beta + 4\gamma$.

De las dos últimas: $P(2) - P(0) = 2\beta + 4\gamma = f(2) - f(0)$ y $\beta + 4\gamma = f'(2)$, restando: $\beta = f(2) - f(0) - f'(2) \cdot 2 + 2f'(2)$... Mejor: $2\beta + 4\gamma = f(2) - f(0)$ y $\beta + 4\gamma = f'(2)$, restando: $\beta = f(2) - f(0) - f'(2)$, luego $\gamma = (f'(2) - \beta)/4 = (f'(2) - f(2) + f(0) + f'(2))/4 = (2f'(2) - f(2) + f(0))/4$.

$$\int_0^\infty e^{-x} P(x)dx = \alpha \int_0^\infty e^{-x} dx + \beta \int_0^\infty x e^{-x} dx + \gamma \int_0^\infty x^2 e^{-x} dx$$
$$= \alpha \cdot 1 + \beta \cdot 1 + \gamma \cdot 2 = \alpha + \beta + 2\gamma.$$

Sustituyendo: $\alpha + \beta + 2\gamma = f(0) + (f(2) - f(0) - f'(2)) + \frac{1}{2}(2f'(2) - f(2) + f(0)) = f(0) + f(2) - f(0) - f'(2) + f'(2) - \frac{1}{2} f(2) + \frac{1}{2} f(0) = \frac{1}{2} f(0) + \frac{1}{2} f(2)$.

$$\boxed{\int_0^\infty e^{-x} P(x)dx = \tfrac{1}{2} f(0) + \tfrac{1}{2} f(2),}$$

coincidiendo exactamente con la fórmula de (a). Esto es consistente porque P tiene grado < 2 y la fórmula de (a) es exacta para polinomios de grado ≤ 2.

ENUNCIADO — PROBLEMA 7

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   $$f(x) - P_2(x) = \frac{1}{6}(x - x_0)(x - x_1)(x - x_2)f'''(\xi(x)), \qquad x_0 \le x \le x_2.$$

   (b) Find the function $\xi(x)$ explicitly for $f(x) = \dfrac{1}{x}$ with $x_0 = 1, x_1 = 2, x_2 = 3$ and $x \ne x_k$ for $k = 0, 1, 2$.

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   $$|y(1) - y_N| \le \frac{Mhe^{b+c}}{2(b+c)}.$$

SOLUCIÓN — PROBLEMA 7

Problema 7 — Error global del método de Euler para EDO no lineal

$y' = f(t, y) = a + by + c\sin y$. El gradiente en y: $\partial f/\partial y = b + c\cos y$, así que $|\partial f/\partial y| \leq b + c$ (constante de Lipschitz $L = b + c$).

El error $e_k = |y_k - y(t_k)|$ satisface la recursión (por Taylor y Lipschitz):

$$e_{k+1} \leq (1 + h(b+c))e_k + \frac{h^2}{2}M, \quad e_0 = 0.$$

Resolviendo la recursión por inducción (serie geométrica):

$$e_k \leq \frac{h^2M/2}{h(b+c)}\left[(1 + h(b+c))^k - 1\right] = \frac{hM}{2(b+c)}\left[(1 + h(b+c))^k - 1\right].$$

En $k = N$ con $Nh = 1$: usando $1 + z \leq e^z$, $(1 + h(b+c))^N \leq e^{Nh(b+c)} = e^{b+c}$.

$$\boxed{|y(1) - y_N| \leq \frac{hM\,e^{b+c}}{2(b+c)}. \blacksquare}$$